Digital Interferometric Open Loop Phase Recovery in a Fiber Optic Gyroscope

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***Abstract*—We present the first use of a Digitally Enhanced Homodyne readout on a Sagnac interferometer, demonstrating the technique's capability as a readout method for high dynamic range, open loop interferometric fiber optic gyroscopes. We achieve an optical phase noise floor of 0.7 μrad/$\sqrt{\text{Hz}}$, and phase linearity of 8 ppm demonstrated over a phase dynamic range of 16π. We rotation calibrate the system using a rate table, confirming an angular random walk (ARW) of 0.0014 deg/$\sqrt{\text{hr}}$, bias instability of 0.012 deg/hr.**

***Index Terms*—Interferometry, modulation, fiber optic gyroscope.**

I. Introduction

INTERFEROMETRIC fiber optic gyroscopes (FOGs) are essential devices in inertial navigation systems [1]. They interferometrically measure the Sagnac phase shift of a closed optical path under rotation and use it to obtain a measurement of rotation rate. The rotation rate can then be integrated to track angular rotation, which forms the basis for the orientation measurement of inertial navigation systems.

One of the main considerations in the application of the Sagnac interferometer as a fiber optic gyroscope is the inherently nonlinear, raised cosine response of the interferometer. This limits the dynamic range, as the phase signal at low rotation rate approaches zero. In most commercial fiber optic gyroscopes, this limitation is overcome by using a closed loop method to dynamically zero out the rotation phase shift using digital phase steps or serrodyne modulation. This method effectively converts the phase response of the interferometer to a sinusoid that is biased around the origin, yielding a linear signal around zero rotation rate [2]. With modern closed loop techniques such as double closed loop modulation [3], a rotation rate dynamic range of up to 2000 deg/s [4] can be achieved.

However, closed loop methods add significant additional design considerations. These include precise control of the timing and gain in the closed loop electronics [5]. The square wave pulses required for closed loop methods can also lead to spurious spikes in the optical power received during the square wave resets. These are typically removed with either optical or electronic gating [1]. Nevertheless, this adds another layer of complexity to the fiber optic gyroscope system. Finally, closed loop methods may suffer from an effect known as lock-in or deadband [6], whereby the bias signal becomes coupled to the serrodyne phase ramp used to null out the signal. This lock-in effect leads to a dead zone, or zero signal, which is observed at low rotation rates. In order to mitigate this effect, parasitic optical and electronic resonances must be adequately suppressed. This includes electronic shielding, minimization of back-reflections and adequate polarization extinction. All of these additional considerations add complexity in attaining high-performance gyroscopic performance. It means there is an ongoing engineering effort to miniaturize, organize and simplify the fiber optic gyroscope to reduce size, weight and power (SWaP) while minimizing performance impact [7].

An alternative to conventional closed loop digital phase extraction schemes are open loop fiber optic gyroscopes. Traditionally, in an open loop scheme, a phase modulator within the Sagnac loop is used to apply phase modulation, however, slower piezo-electric (PZT) fiber stretchers are commonly used instead of electro-optic devices [8]. Following photodetection, the odd and even harmonics are demodulated to recover the Sagnac phase [9]. However, these methods are limited in scale factor accuracy and dynamic range by the precision of the gain control of the PZT modulator [10].

Digital Interferometry (DI) is a promising open loop interferometric phase extraction scheme that has previously demonstrated phase sensitivities of ~ 10 μrad/$\sqrt{\text{Hz}}$ along with a continuous and linear, unwrapped phase readout [11], [12]. A demonstrated capability of the technique is its ability to resolve phase dynamics over an extended dynamic range over many cycles of optical phase [13]. The technique works by introducing a pseudo-random noise (PRN) phase modulation which results in an optical spread-spectrum. We note that PRN phase modulation has been used to broaden fiber optic gyroscope light sources [14], [15], however, DI extends beyond this to use the modulation to pseudo-randomly sample and reconstruct the optical interference fringe. This ability, along with the time-of-flight correlation properties of PRN modulation, enables DI to achieve an open loop phase readout with the additional ability to time-of-flight gate spurious signals.

Here, we demonstrate and characterise an open loop, linear, large dynamic range readout using a subset of digital interferometry, named Digitally Enhanced Homodyne Interferometry (DEHoI) [16]. The use of DEHoI enables us to retain a baseband

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readout required for a reciprocal Sagnac interferometer and enables an open loop phase readout that can extend over many multiples of π, which is traditionally only possible using double closed loop modulation techniques. In addition to its intrinsic open loop readout, it also affords the flexibility to adapt the modulation frequency and measurement bandwidth independent of the gyroscope coil transit frequency. This absolves the proposed method from the common challenge of synchronising modulation speeds to the coil transit frequency to achieve optimal gyroscopic performance [17]. We demonstrate this modulation scheme on an an otherwise canonical fiber optic gyroscope.

II. Digital Phase Extraction Scheme

Digital interferometry uses pseudo-random phase modulation to encode the optical phase. This allows for the propagation time of the pseudo-random code through an optical system to be identified by the delay of the code's auto-correlation in demodulation. Critically, this auto-correlation is low-valued for all time delays except for when it is an integer multiple of the code period. In DEHoI specifically, two pseudo-random binary codes are used to create a 4-level phase constellation, akin to Quadrature Phase Shift Keying (QPSK) modulation. These 4 levels allow for discrete sampling of each quadrature of the IQ plane, with each code corresponding to a specific quadrature. We can describe the QPSK modulated electric field in terms of its constituent pseudo-random binary codes using the following expression:

$$E(t) = \tilde{E}(t)M(t)e^{i\omega t + i\phi(t)} \tag{1}$$

$$M(t) \equiv C_I(t) + iC_Q(t) \tag{2}$$

where $C_I(t)$ and $C_Q(t)$ are mappings of the I and Q binary codes to ± 1. We write this in compact form as $M(t)$. $\tilde{E}(t)$ represents the time varying amplitude of the field, ω the optical carrier frequency and $\phi(t)$ a time varying phase. The code delay, τ is the propagation time from the point of modulation to the point of decoding in signal processing. This is typically dominated by the time-of-flight through the optical system. When implemented in a two beam interferometer this results in two optical fields superposed at the photodetector, each encoded with a uniquely delayed QPSK code.

In this paper we demonstrate this open loop DEHoI phase readout in a Sagnac interferometer architecture interrogated by a SLD source. To implement a digital interferometric readout, we require the phase modulation $M(t)$ to be encoded on both the clockwise (CW) and counterclockwise (CCW) metrology fields. The modulation is applied to the fields within the coil, as due to the symmetric nature of the Sagnac interferometer, any modulation that is encoded outside the interferometer is common, and does not manifest itself in the measurement. While this reciprocity condition is advantageous for suppressing optical source frequency noise and other common noise sources, it also requires any modulation for readout purposes to be applied within the sensing coil.

Similarly to canonical FOG architecture, we use a Multifunction Integrated Optical Chip (MIOC), a solid state optical device that integrates a Y-junction coupler with dual Electro-optic modulators (EOM) within the Sagnac coil, as shown in Fig. 1.

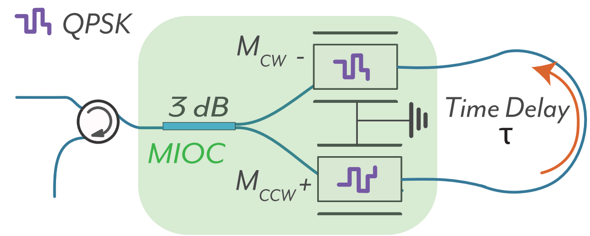

Fig. 1. By using an MIOC in a push-pull configuration, the QPSK modulation results in a pair of codes inverted with respect to each other. The clockwise and counterclockwise fields are thus doubly modulated with a prompt QPSK code, and an inverted code delayed by the coil transit time τ with respect to the first modulator. Using DEHoI to decode the resulting doubly modulated field then recovers the interference between the CW and CCW fields, enabling a measurement of the Sagnac phase.

These modulators are operated in a push-pull configuration, such that the phase shift imparted by one modulator is the negative of the other.

To determine the encoding combination achieved by the modulators, we follow the propagation of the metrology fields through the sensing coil in both the CW and CCW directions. Looking first at the counterclockwise field, From Fig. 1 we see that the field is first modulated by the modulator M_{CCW}. This introduces a τ delay to the code imparted by this modulator relative to the CW modulator M_{CW}, where τ is the coil transit time. By repeating the analysis for the field in the clockwise direction, we may then write the clockwise and counterclockwise fields as the following:

$$E_{CCW}(t) = \tilde{E}(t)\left[M_{CW}(t) + M_{CCW}(t-\tau)\right]e^{i\omega t - i\phi_{CCW}(t)}, \tag{3}$$

$$E_{CW}(t) = \tilde{E}(t)\left[M_{CW}(t-\tau) + M_{CCW}(t)\right]e^{i\omega t + i\phi_{CW}(t)}. \tag{4}$$

The push-pull modulation can be expressed mathematically as $M_{\mathrm{CCW}} = -M^*_{\mathrm{CW}}$. Thus, we may rewrite (3) and (4) in terms of a common modulation $M(t)$:

$$E_{CCW}(t) = \tilde{E}(t)\left[M(t) - M^*(t-\tau)\right]e^{i\omega t + i\phi_{CCW}(t)}, \tag{5}$$

$$E_{CW}(t) = \tilde{E}(t)\left[M(t-\tau) - M^*(t)\right]e^{i\omega t + i\phi_{CW}(t)}. \tag{6}$$

At the photodetector, we obtain the superposition of the clockwise and counterclockwise fields:

$$P_{\mathrm{D}}(t) = \text{D.C. terms} + \underbrace{E_{\mathrm{CCW}}E^*_{\mathrm{CW}} + E^*_{\mathrm{CCW}}E_{\mathrm{CW}}}_{\text{interference terms}}, \tag{7}$$

Using (2), we expand the interference term by rewriting the compound codes $M(t)$ in the form of their constituent binary I and Q codes.

$$\begin{aligned} P_D = &|\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|C_I(t)C_I(t-\tau)\cos(\Delta\phi(t)) \quad (II)\\ &+ |\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|C_I(t)C_Q(t-\tau)\sin(\Delta\phi(t)) \quad (IQ)\\ &- |\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|C_Q(t)C_I(t-\tau)\sin(\Delta\phi(t)) \quad (QI)\\ &+ |\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|C_Q(t)C_Q(t-\tau)\cos(\Delta\phi(t)) \quad (QQ)\\ &+ \text{D.C. Terms} \end{aligned} \tag{8}$$

We obtain four terms encoding the Sagnac phase, each multiplied by a combination of I and Q codes. These four terms are decoded using four parallel decoders on an field programmable gate array (FPGA). Each decoder performs a code correlation [16] with appropriately delayed local copies of the codes for the corresponding interference term. This is done by multiplying the photodetector signal by the decoding codes, and integrating over a number of code symbols, N, where N is the QPSK code length. By decoding at the appropriate delays, we can selectively ensure the code correlation, R, is large for only the desired signals. Signals which are differently delayed will have a suppressed code correlation of only R/N. This effectively allows for the isolation of each phase shifted component, yielding the four decoded outputs shown in (9).

$$\begin{aligned} II &= |\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|R_{II}\cos(\Delta\phi(t)) \\ IQ &= |\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|R_{IQ}\sin(\Delta\phi(t)) \\ QI &= |\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|R_{QI}\sin(\Delta\phi(t)) \\ QQ &= |\tilde{E}_{cw}(t)\tilde{E}_{ccw}(t)|R_{QQ}\cos(\Delta\phi(t)) \end{aligned} \tag{9}$$

Following demodulation we can then use a linear combination of these four measurements to construct an IQ projection for the actual measurement phasor, as shown in (10):

$$s(t) = \underbrace{(II + QQ)}_{\text{In-Phase (I)}} + \underbrace{i(IQ - QI)}_{\text{Quadrature (Q)}} \tag{10}$$

The phase difference $\Delta\phi(t)$ is then given by:

$$\Delta\phi(t) = \tan^{-1}\left(\frac{\text{Im}\{s(t)\}}{\text{Re}\{s(t)\}}\right) \tag{11}$$

At this stage the readout is defined over the full unit-circle. Extending beyond a cycle can be achieved using a standard 1D phase unwrapping algorithm, such as the MATLAB unwrap function, or equivalent on FPGA. With this method we are able to reconstruct the Sagnac phase with an open loop, continuous, and high dynamic range readout. We note that the decoding architecture outlined here is identical to prior implementations of DEHoI [18], [19]. The calibration for optical phase to rotation rate (Ω) is then given by the standard Sagnac equation, shown in (12), which depends on the coil length (L), coil diameter (d) and source wavelength (λ):

$$\Omega(t) = \frac{\lambda c}{2\pi L d}\Delta\phi(t) \tag{12}$$

III. Experimental Demonstration

We use a Thorlabs (S5FC1005S) benchtop superluminescent diode source with a spectral width of 50 nm centered around 1550 nm. The PM fiber coupled output of this source is sent through a circulator and 3 dB splitter, contained on an Multi-function Integrated Optical Chip - MIOC, into the Sagnac coil as shown in Fig. 2. The PM fiber coil, MIOC and circulator were provided by Advanced Navigation for the project. Photodetection was performed using an Insight Photonics BPD1-400 balanced detector configured to operate in single detector mode.

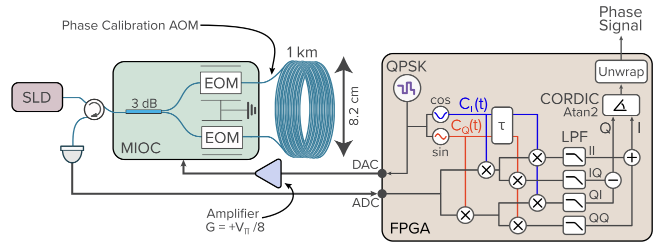

Fig. 2. The experimental setup consists of an MIOC in a Sagnac interferometer with a polarization-maintaining sensing coil of length 1 km long and diameter of 8.2 cm. We use the MIOC in a push-pull configuration to modulate the QPSK code which was generated on FPGA. The received optical signal is detected and digitize prior to demodulation with prompt and coil delayed QPSK codes according to their in-phase (cosine) and quadrature (sine) projections. Averaging (LPF) obtains the code correlation and outputs IQ data to a CORDIC based Arctan2 operation and then threshold detection unwrapper to compute the Sagnac phase signal. For calibrating the phase readout, an acousto-optic modulator (AOM) is added asymmetrically to the coil. This is removed for noise and rotation tests.

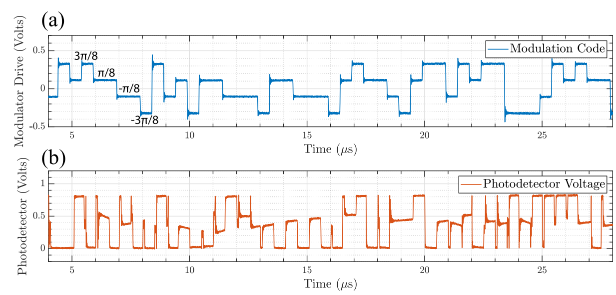

Fig. 3. Fig. 3(a) shows an oscilloscope measurement of the 2 MHz, 4 level QPSK code generated by the DAC on the FPGA. The multiples of $\pi/8$ designate the EOM modulation depth corresponding to the voltage levels. Fig. 3(b) shows an oscilloscope measurement of the voltage output of the photodetector. This is proportional to the expression for optical power at the photodetector given in (8).

Our DEHoI scheme uses an MIOC to phase encode the two interferometer fields with a 4-level QPSK code. For the modulation, we use an 11-bit linear feedback shift register to generate the m-sequences, using a self-delay method to generate two orthogonal code pairs. The use of m-sequences as the generator codes therefore results in a QPSK modulation with a code length of 2047 chips (symbols). This is modulated onto the CW and CCW fields at a 2 MHz chip frequency, corresponding to a physical chip length in optical fiber of ∼100 m. A sample of the QPSK code used is shown in Fig. 3(a). Since the DeHoI readout requires averaging over a code length to recover the necessary autocorrelation, the final decimated sampling frequency of 977 Hz determines the readout bandwidth.

The generation and decoding of the QPSK code are handled on a field programmable gate array (FPGA). In addition to the code, the FPGA adds further options for phase and frequency modulation via an acousto-optic modulator for calibration purposes, as discussed in the following section. The decoding process consists of re-orienting the phase constellation with respect to the code pair that has been detected. This is done in two stages. The first stage corresponds to the prompt code, here given a delay of zero. We compute the IQ projection of the prompt code, with a multiplication operation to obtain the IQ components of the optical signal. This is then passed to the second stage, which repeats the process for the delayed code, here with a delay of τ. A 2nd order cascaded integrator comb low pass filter is used to integrate over the full m-sequence code length, recovering the code correlation for each of the four possible combinations. This yields the vector components described in (9). The components are linearly combined according to (10) to obtain the final IQ pair for the interferometer phase.

The decoded IQ pair is presented to a coordinate rotation digital computer (CORDIC) algorithm (Xilinx CORDIC IP Core) to compute the wrapped phase which spans from $(-\pi, \pi]$. A threshold detection phase unwrap operation then checks and remaps discontinuities in the wrapped phase corresponding to 2π phase jumps, outputting the unwrapped phase for data acquisition. For initial phase calibration tests, the CORDIC operation was bypassed with raw IQ data being used to compute the unwrapped phase in post-processing. We saw no degradation in performance between the two methods.

For the FPGA, we use a National Instruments, Virtex 5, 7965R FPGA with the analog interface provided by a NI 5781 transceiver. The transceiver operates at 100 MS/s on both the two input ADCs and two output DACs. This provides the I/O card with an effective 40 MHz bandwidth, capable of generating the necessary modulation for the MIOC. After being output from the DAC, the MIOC drive signal is amplified using a Mini-Circuits ZHL-32 A RF amplifier and sent to the MIOC (YOEC GC15YB4010).

The MIOC allows for push-pull modulation of the QPSK code. This means we are able to modulate the counter-propagating beams with a differentially delayed QPSK code. For the CW beam, the code is therefore delayed by τ, where τ is the coil transit time, with respect to the CCW path. The phase difference between the CW and CCW paths is encoded in the photodetector output, as shown in Fig. 3(b), then coherently extracted by decoding for the CW-CCW code delay combinations shown in (8).

IV. DI Phase Readout Linearity

The preliminary measurement of the Sagnac interferometer was to confirm the phase calibration and linearity of the phase readout. To test the phase readout linearity, we insert an 80 MHz acousto-optic modulator (AOM) (IntraAction FCM-801E5A) into one end of the 1 km coil path. The AOM RF frequency shift is then linearly tuned over a range of 1.6 MHz using a sawtooth waveform, all of which were generated on FPGA and output via a second DAC. The tuning changes the differential optical phase accrued between the two counter-propagating beams, generating a linear phase ramp readout. Since the coil length of 1 km corresponds to a free spectral range of 200 kHz, the phase ramp scans over 8 free spectral ranges and is equivalent to 16π of phase shift.

In Fig. 4(a) we see that the resulting phase ramp tracks over the expected 16π or 50 radians phase shift. To compute the linearity of the phase readout, a linear fit is performed on the phase readout, and the residuals are plotted in Fig. 4(b). The standard deviation of the residuals is computed and divided by the total phase excursion of 50 radians, yielding a phase readout non-linearity of 8ppm.

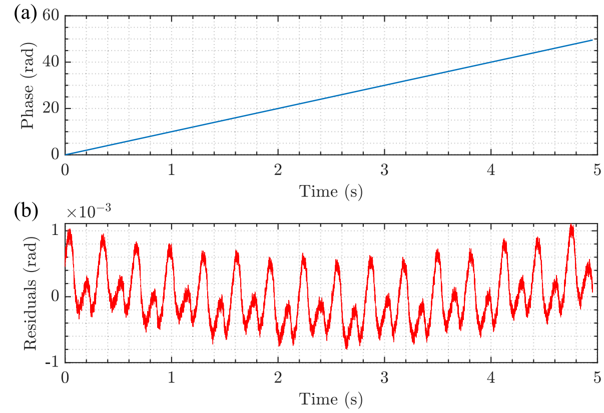

Fig. 4. Fig. 4(a) shows the interferometer phase output as the AOM RF frequency shift is tuned over 1.6 MHz, corresponding to a 50 rad phase shift. Fig. 4(b) shows the residuals from a linear fit of the phase output. The standard deviation of the residuals is divided by the phase output amplitude to obtain a non-linearity figure of 8 ppm.

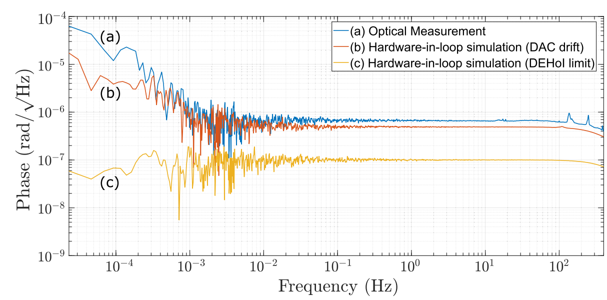

Fig. 5. A frequency domain plot of the optical phase sensitivity, trace (a), for the DI Sagnac interferometer is shown with a high frequency noise floor of 0.7 μrad/$\sqrt{\text{Hz}}$. In trace (b), the modulation code is sent out from the DAC on the FPGA, and back into the ADC on the FPGA for demodulation, allowing measurement of DAC drift. The low frequency roll up from 2 mHz corresponds with DAC gain drift which is on the timescale of laboratory air conditioner cycles. Trace (c) shows the result of a hardware-in-loop simulation which isolates the noise limit of the DEHoI technique. This is well below the optical measurement in Trace (a).

V. Noise Performance

Following calibration, we proceed to characterize the phase noise performance of the readout. To ensure optimal noise performance, the AOM used for phase readout linearity measurements was removed, and the Sagnac interferometer enclosed in a shroud to minimise effects of air currents. A noise measurement was taken on an optical table, with the optics bolted to the table surface. The measurements were taken overnight for 12 hours to minimise personnel activity in the laboratory.

In trace (a) of Fig. 5, we plot the amplitude spectral density computed from a time series measurement of the optical system, showing the phase sensitivity of the Sagnac interferometer phase readout of 0.7 μrad/$\sqrt{\text{Hz}}$ above approximately 1 mHz. The predicted shot noise limited phase sensitivity for this measurement is 0.15 μrad/$\sqrt{\text{Hz}}$. Analog front end electronic noise for our current system was evaluated to be at approximately 0.2

μrad/$\sqrt{\text{Hz}}$. Given no relative intensity noise (RIN) mitigation in this system, the excess noise of the SLD is hypothesised to be the cause of the white noise floor level. At frequencies below approximately 2 mHz, the noise floor is seen to roll up, which we hypothesize is due to thermal drift in the system.

To isolate the source of this drift, we conduct two hardware simulation measurements. This used the same FPGA hardware used in the experiment to generate a simulated photodetector signal. The first measurement isolates the performance of the DAC by driving the ADC with the QPSK modulation signal, as shown in Fig. 3(a). This is digitized by the ADC and the simulated photodetector output is computed. The simulated output is passed to the demodulation signal processing, recovering the simulated phase. The resulting frequency domain noise floor is shown in the trace (b) of Fig. 5. We note the high frequency noise floor is higher than the ADC noise floor, as the DAC was operating at the voltage optimised for the optical modulator and not the ADC. Importantly, the noise floor roll-up between 0.2 mHz and 1 mHz corresponds with that of the optical measurement. This indicates that the source of drift here is most likely temperature dependent drifts in the DAC gain voltage used to drive the MIOC, which matches the timescale of laboratory air conditioner cycles. This would result in fluctuations in the modulation depth achieved from the MIOC, which manifest as phase drift. This is consistent with conventional fiber optic gyroscope systems which typically require stabilization of the modulation signal gain to obtain long term stability [1].

In trace (c) of Fig. 5, a frequency domain noise plot of a hardware-digital-in-loop simulation is shown. This computes the simulated photodetector signal internally, isolating the noise limit of the DEHoI modulation. The resulting noise floor of 0.1 μrad/$\sqrt{\text{Hz}}$ demonstrates that optical phase measurement is not limited by the performance of DEHoI, including at low Fourier frequencies below 1 mHz.

VI. Rotation Calibration

To confirm operation as a gyroscope, the final experiment was to conduct a direct rotation measurement. To do this, the system was mounted onto an Acutronics AC1120Si V2 rate table. Due to space constraints, the FPGA was replaced during this process with the NI 7935R partnered with the NI5782 transceiver module with a 125 MS/s digital sampling period. The FPGA code was ported to the new platform with no functional changes, and all the modulation rates were maintained at the original level. A WIFI router was mounted on the stage and connected to the FPGA controller for data transfer and control.

We recorded a series of time series phase measurements at rotation rate increments between -360 and 360 deg/s to directly measure the gyroscope scale factor, seen in Fig. 6(a). Phase values correspond to the mean of 30 s measurements at each rotation rate recorded. The entire dataset was obtained within a single two hour period. A linear trendline was fitted in Fig. 6(a) to determine the gyroscope scale factor and the residuals shown in Fig. 6(b). The measured scale factor of the system determine to be 50.25 deg/s/rad. Using the measured scale factor and demonstrated phase sensing range in Fig. 4, we

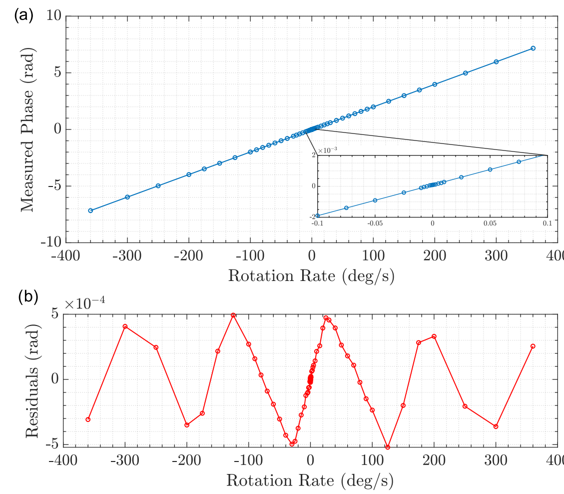

Fig. 6. Rotation calibration was conducted on a Acutronics AC1120Si V2 rate table using the same optical system as the benchtop tests. a) demonstrates the scale factor linearity in excess of $\pm\pi$ radians phase excursion. The inset shows the ability to resolve rotation rates near zero. b) displays the residual from the linear fit to the data presented in a). We obtain a bounded residual of 1 mrad peak to peak, which is in agreement with the phase readout linearity measurements shown in Fig. 4.

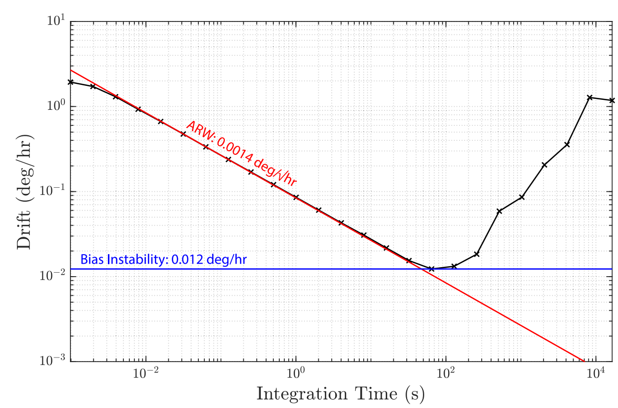

Fig. 7. An Allan deviation of the time domain measurement calibrated to equivalent rotation rate. The angular random walk (ARW) reaches 0.0014 deg/$\sqrt{\text{hr}}$, while the bias instability of 0.012deg/hr is achieved at 64 seconds.

can expect that the rotation rate sensing dynamic range for this architecture can be as high as $\pm$2512.5 deg/s. As the only strict limit to phase dynamic range is the bit-width of the unwrapped phase data, we can expect the theoretical maximum angular rate range to be several orders of magnitude larger than this value.

We confirm the DI Sagnac interferometer produces a linear phase response with rotation across a greater than 4π range which corroborates phase linearity measurements shown in Fig. 4, with the non-linearity also being bounded at the same 1 mrad scale. The inset in Fig. 6(a) shows the rotation rate measurements between -0.1 and 0.1deg/s which demonstrates this system's sensitivity to rotation rates around 0 deg/s. Given the repeatable and bounded nature of the residual, it may be possible to remove this effect via calibration in a production system to further improve the residual scale-factor non-linearity.

In Fig. 7, we plot the Allan deviation from the measurement shown in Fig. 5, calibrated to rotation rate. The calibration from phase to rotation rate was computed using the scale factor computed from the rate table data in Fig. 6. The calibrated angular random walk from the Allan deviation is 0.0014 deg/$\sqrt{\mathrm{hr}}$, with the bias instability measured to be 0.012 deg/hr at 64 seconds integration time. Both these values approach the inertial navigation grade requirement for bias instability (0.01 deg/hr) and exceeds the requirement for angular random walk (0.002 deg/$\sqrt{\mathrm{hr}}$) [1], [20]. We expect that the bias instability can be improved by lengthening the integration time using temperature calibration [21] as well as by stabilizing the reference voltage of the DACs generating the MIOC modulation signal [22], both of which are common practice in commercial systems.

Comparing gyroscope performance grades, the prototype system meets tactical grade performance. Gyroscope products in this area, such as the iXBlue UmiX U9 and Honeywell HG2800 have comparable dynamic range ($\sim \pm 2000$ deg/s) and bias instability ($\sim$0.02 deg/hr). Our prototype system ARW however is significantly improved ($\sim 10\times$), indicating it can resolve high-speed dynamics to a higher precision.

This is contrasted with navigation and strategic grade solutions which do have superior ARW and bias instability (eg. Advanced Navigation D90, iXBlue ASTIX 1120, ASTIX 200) however with a significant impact to dynamic range, limiting them to $<< \pm$ 1000 deg/s. This largely arises from the aforementioned double closed-loop operation [3]. The prototype system therefore lands between these two performance regimes, and is particularly suited to precision high-dynamic range use-cases.

VII. Conclusion

We have demonstrated the use of digitally enhanced homodyne interferometry (DEHoI) to obtain an open loop phase readout from a Sagnac interferometer, for application in interferometric fiber optic gyroscopes. We confirm the gyroscope scale factor via rotation rate measurements, and obtain a bias instability of 0.012 deg/hr and an angular random walk of 0.0014 deg/$\sqrt{\mathrm{hr}}$. Furthermore, the digital interferometric readout scheme enables us to obtain a phase readout non-linearity of 8 ppm over a dynamic range of 16π with an open loop readout, well beyond the phase wrap threshold of $\pm\pi$ radians. The ability to measure rotation rates beyond the wrap point was confirmed by rate table measurements.

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